PYQ 2010–2025

WBUHS BIOCHEMISTRY · MODULE 03

Lipid Metabolism

β-Oxidation · Ketone Bodies · Cholesterol · Lipoproteins · Fatty Acid Synthesis

SUBJECT	EXAM RELEVANCE	PYQ FREQUENCY
Biochemistry — Paper 1	Group A · B · C · D	10+ times (2010–2025)

⭐ **APPEARED IN WBUHS — YEARS**

2010 · 2010-S · 2011-S · 2012-S · 2013 · 2013-S · 2014 · 2015 · 2015-S

2017 · 2017-S · 2018 · 2019 · 2019-S · 2021 · 2022 · 2023 · 2024 · 2025

01 The Big Picture

WHY THIS MATTERS CLINICALLY

Lipid metabolism is where fat becomes energy, where drugs like **statins** intervene, and where diseases like **atherosclerosis, fatty liver, DKA, and familial hypercholesterolaemia** originate. The examiner loves lipid questions because they demand pathway knowledge, enzyme names, and clinical application all at once — often in a case-based Group A.

// MASTER OVERVIEW — FATE OF LIPIDS

Dietary Lipids
(TAG, Cholesterol, PL)

↓

Digestion + Absorption (Bile salts, Pancreatic lipase) → Chylomicrons (Lymph → Blood)

↓

Fatty Acids → β-Oxidation (Mitochondria) → Acetyl-CoA → TCA → ATP

↓ (if excess)

Ketone Bodies (in starvation / DM) ⟷ Cholesterol Synthesis (HMG-CoA pathway)

02 β-Oxidation of Fatty Acids ★ EXAMINER FAV

📌 **ONE-LINER**

Sequential removal of **2-carbon units** as Acetyl-CoA from the β-carbon of a fatty acid, occurring in the **mitochondrial matrix**, generating $FADH_2$ + NADH per cycle.

// CARNITINE SHUTTLE — ENTRY INTO MITOCHONDRIA

Fatty Acid (cytoplasm) → Acyl-CoA Synthetase (ATP) → Fatty Acyl-CoA

↓ CPT-I RLS (outer membrane)

Acyl-Carnitine → Translocase (inner membrane) → Acyl-Carnitine (matrix)

↓ CPT-II (inner membrane)

Fatty Acyl-CoA (matrix) → β-Oxidation ✓

⭐ **CPT-I is the rate-limiting step** RLS of β-oxidation. Inhibited by **Malonyl-CoA** (the first committed intermediate of fatty acid synthesis) — this is how the cell prevents simultaneous synthesis and breakdown of fatty acids.

// ONE ROUND OF Β-OXIDATION (PALMITATE C16 EXAMPLE)

Palmitoyl-CoA (C16)

↓ Step 1: Acyl-CoA Dehydrogenase → $FADH_2$

Trans-Δ^2-Enoyl-CoA

↓ Step 2: Enoyl-CoA Hydratase → H_2O added

L-3-Hydroxyacyl-CoA

↓ Step 3: Hydroxyacyl-CoA Dehydrogenase → NADH

3-Ketoacyl-CoA

↓ Step 4: Thiolase → CoA added

Acetyl-CoA (C2) ✓ + Acyl-CoA (C14) → next round

Palmitate (C16): 7 rounds of β-oxidation 8 Acetyl-CoA produced	**Per round:** 1 $FADH_2$ + 1 NADH = 4 ATP	**Net ATP (C16):** 106 ATP total (−2 for activation)

SPECIAL OXIDATIONS

α **α-Oxidation:** Oxidation at α-carbon. For branched-chain fatty acids (e.g. phytanic acid). Requires Phytanoyl-CoA hydroxylase. Defect → **Refsum's disease**.

ω **ω-Oxidation:** Oxidation at terminal (ω) carbon. Minor pathway in ER. Produces dicarboxylic acids. Important in MCAD deficiency.

PX **Peroxisomal β-Oxidation:** For very long chain FA (>C20). Generates H_2O_2 (catalase destroys it). No ATP produced directly. Defect → **Zellweger syndrome**, **Adrenoleukodystrophy (ALD)**.

03 # Ketone Bodies ★ EXAMINER FAV

// KETOGENESIS — SYNTHESIS (IN LIVER MITOCHONDRIA)

2 × Acetyl-CoA → Thiolase Acetoacetyl-CoA

↓ + Acetyl-CoA → HMG-CoA Synthase RLS

HMG-CoA

↓ HMG-CoA Lyase

Acetoacetate

↙ ↓

β-Hydroxybutyrate
(NADH reduction)

Acetone
(spontaneous decarboxylation)
→ *fruity breath*

🏭 **Ketone bodies are made in LIVER but used in EXTRAHEPATIC tissues** (brain, heart, muscle, kidney). Liver lacks Succinyl-CoA transferase (thiophorase) — so it cannot use its own ketones.

🔄 **Ketolysis (utilisation) — in extrahepatic tissues:** β-Hydroxybutyrate → Acetoacetate (by HBD) → Acetoacetyl-CoA (by Succinyl-CoA transferase, needs Succinyl-CoA) → 2 Acetyl-CoA → TCA → ATP.

🩺 **KETOSIS — CLINICAL SCENARIOS**

- **Starvation ketosis:** Mild — OAA diverted to GNG → Acetyl-CoA cannot enter TCA → ketogenesis. Controlled, compensated. pH usually normal.
- **DKA (Diabetic Ketoacidosis):** Severe — No insulin → uncontrolled lipolysis → massive FFA → overwhelming ketogenesis. Acetone breath, pH < 7.3, High Anion Gap Metabolic Acidosis. Emergency.
- **Alcoholic ketoacidosis:** Ethanol → NADH ↑ → OAA → Malate (depletes OAA) → same result as starvation.
- **Key exam trap:** Both starvation AND DKA cause ketosis but **DKA is more severe** because insulin deficiency removes the brake on lipolysis completely.

04 Fatty Acid Synthesis (Lipogenesis)

// DE NOVO FATTY ACID SYNTHESIS — KEY STEPS

Acetyl-CoA (mitochondria) → Citrate shuttle to cytoplasm → Acetyl-CoA (cytoplasm)

↓ Acetyl-CoA Carboxylase (ACC) RLS + Biotin + CO_2 + ATP

Malonyl-CoA (3C)

↓ Fatty Acid Synthase Complex (FAS) — 7 rounds × 2C addition

Palmitate (C16:0) — first product

Location: Cytoplasm (+ SER for elongation)	**Reducing agent:** NADPH (from HMP shunt + malic enzyme)	**Cost per Palmitate:** 7 ATP + 14 NADPH + 8 Acetyl-CoA

⭐ **ACC (Acetyl-CoA Carboxylase)** RLS — Activated by: Citrate, Insulin. Inhibited by: Palmitoyl-CoA (product inhibition), Glucagon, Epinephrine (via phosphorylation by PKA). **Malonyl-CoA** — the product — also inhibits CPT-I, preventing simultaneous FA synthesis + breakdown.

SYNTHESIS VS β-OXIDATION

Location: **Cytoplasm** vs **Mitochondria**

Carrier: **ACP (Synthesis)** vs **CoA (Oxidation)**

NADPH consumed vs NADH+$FADH_2$ produced

Citrate shuttle IN vs Carnitine shuttle IN

Stereochemistry: **L** vs **D** intermediate

ELONGATION & DESATURATION

Elongation: ER + mitochondria (Malonyl-CoA units)

Desaturation: ER → Δ9-desaturase (most common)

Humans: **cannot introduce double bonds beyond C9**

∴ Linoleic (ω6) + α-Linolenic (ω3) = **Essential FA**

EFA deficiency → scaly skin, ↑ susceptibility to infection

05 Cholesterol Synthesis ★ EXAMINER FAV

// CHOLESTEROL SYNTHESIS — KEY STEPS (EXAM CONDENSED)

3 Acetyl-CoA → HMG-CoA Synthase (cytoplasm) HMG-CoA

↓ HMG-CoA Reductase RLS + 2 NADPH

Mevalonate (6C) ← Statin target

↓ → Isopentenyl pyrophosphate (IPP, 5C) → Squalene (30C)

Lanosterol → 19 steps **Cholesterol (27C)**

Location: Cytoplasm + ER (mainly liver)	**ACAT:** Esterifies cholesterol for storage	**LCAT:** Esterifies cholesterol in plasma (HDL)

HMG-CoA Reductase RLS — Inhibited by: Statins (competitive), High intracellular cholesterol (via SREBP-2 pathway), Glucagon. Activated by: Insulin. **Statins** (Atorvastatin, Rosuvastatin) → ↓ cholesterol → upregulate LDL receptors → ↑ LDL clearance from blood.

PRODUCTS OF CHOLESTEROL

1. **Bile acids** (primary: cholic, chenodeoxycholic) — essential for fat digestion
2. **Steroid hormones** — glucocorticoids, mineralocorticoids, sex hormones
3. **Vitamin D** — 7-dehydrocholesterol → D3 (skin, UV light)
4. **Cell membrane** — modulates fluidity

06 Plasma Lipoproteins ★ EXAMINER FAV

LIPOPROTEIN	ORIGIN	MAIN LIPID	KEY APO	FUNCTION
CHYLOMICRON	Intestine	TAG (85%)	ApoB-48, ApoC-II, ApoE	Transport dietary TAG to tissues
VLDL	Liver	TAG (55%)	ApoB-100, ApoC-II, ApoE	Transport endogenous TAG
IDL	VLDL remnant	TAG + Chol	ApoB-100, ApoE	Intermediate; → LDL
LDL	IDL (VLDL remnant)	Cholesterol (45%)	ApoB-100	**Delivers cholesterol to cells — "Bad"**
HDL	Liver + Intestine	Protein (50%)	ApoA-I, ApoA-II	**Reverse cholesterol transport — "Good"**

// VLDL METABOLISM & REVERSE CHOLESTEROL TRANSPORT

VLDL (from liver) → LPL (capillaries) → Releases TAG → tissues

↓ becomes IDL → LPL again

LDL → LDL receptor (ApoB-100) → Cell takes up cholesterol

HDL (nascent) from liver/intestine → LCAT (ApoA-I activates) → Picks up free cholesterol from tissues

↓

Mature HDL → CETP → Cholesterol → Liver (for excretion as bile)

⚖️ **LPL vs. HSL (Classic Viva Question):** Lipoprotein Lipase (LPL) sits on capillaries, is activated by Insulin and ApoC-II, and clears TAGs from the blood into tissues (Fed state). Hormone Sensitive Lipase (HSL) sits inside adipocytes, is activated by Glucagon/Epinephrine, and breaks down stored fat to release free fatty acids into the blood (Fasting state).

Familial Hypercholesterolaemia (FH): LDL receptor defect (ApoB-100 mutation) → LDL not cleared → ↑ serum LDL → premature atherosclerosis + xanthomas. Heterozygous: LDL ~2× normal. Homozygous: LDL ~6× normal, MI before age 20.

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07 Phospholipids & Glycolipids

KEY PHOSPHOLIPIDS

Lecithin (PC) — most abundant; cell membrane + lung surfactant

Cephalin (PE) — cell membranes; coagulation

Sphingomyelin — myelin sheath; SM:PC ratio used for fetal lung maturity

Cardiolipin — inner mitochondrial membrane; antigen in syphilis test (VDRL)

Plasmalogen — heart + brain; ether phospholipid

PHOSPHOLIPASES

PLA_2 — releases arachidonic acid → eicosanoids (PGs, TXA_2, LTs)

PLC — generates DAG + IP_3 (2nd messengers)

PLD — generates phosphatidic acid

Cobra venom → PLA_2 → haemolysis via lysolecithin

SURFACTANT — CLASSIC PYQ (2015-S, 2019-S)

- **Dipalmitoyl lecithin (DPPC)** = lung surfactant. Produced by Type II pneumocytes. Reduces surface tension in alveoli at end-expiration → prevents alveolar collapse.
- **Respiratory Distress Syndrome (RDS) / Hyaline Membrane Disease:** Premature infants (<35 weeks) — surfactant deficiency → alveolar collapse → respiratory failure. Maternal corticosteroids accelerate surfactant synthesis.
- **L:S ratio** (Lecithin:Sphingomyelin) in amniotic fluid — >2 = mature fetal lungs.

LIPID STORAGE DISEASES (SPHINGOLIPIDOSES) — HIGH YIELD

- **Tay-Sachs Disease:** Hexosaminidase A deficiency → Ganglioside GM2 accumulates → Cherry-red spot on macula, neurodegeneration, no hepatosplenomegaly.
- **Gaucher's Disease:** Glucocerebrosidase deficiency → Glucocerebroside accumulates → Crumpled tissue paper macrophages, hepatosplenomegaly, bone crises. (Most common).
- **Niemann-Pick Disease:** Sphingomyelinase deficiency → Sphingomyelin accumulates → Cherry-red spot on macula + hepatosplenomegaly (foam cells).

08 Eicosanoids

// EICOSANOID SYNTHESIS FROM ARACHIDONIC ACID

Membrane Phospholipids → PLA_2 (activated by stimuli) → **Arachidonic Acid** (C20:4)

↓ COX (Cyclo-oxygenase)

PGH_2

PGE_2, PGI_2 (prostacyclin) → vasodilation

TXA_2 (thromboxane) → vasoconstriction + platelet agg.

↓ LOX (Lipoxygenase)

5-HPETE

Leukotrienes (LTA_4, LTB_4, LTC_4) → bronchoconstriction → inflammation (asthma)

COX inhibitors: Aspirin — irreversible acetylation of COX (serine residue). NSAIDs — reversible competitive inhibition. COX-2 selective inhibitors (Celecoxib) → reduced GI side effects. **Aspirin's anti-platelet effect persists 7–10 days** (platelet lifespan) because platelets have no nucleus — cannot synthesise new COX.

09 Fatty Liver & Lipotropic Factors

DEFINITION

Fatty liver (hepatic steatosis) = accumulation of TAG in hepatocytes >5% of liver weight. Results from imbalance between TAG synthesis and VLDL export.

// CAUSES OF FATTY LIVER

↑ FA delivery to liver (starvation, obesity, DM)	↑ FA synthesis (high carb diet, insulin)	↓ FA oxidation (alcohol, hypoxia)

↓ Apoprotein synthesis (protein malnutrition, CCl_4)	↓ Lipotropic factors (↓ choline, methionine)

↓ All lead to:

TAG accumulates in hepatocytes → Fatty Liver

🧪 **Lipotropic factors** prevent fatty liver by promoting VLDL synthesis/export: **Choline** (for phosphatidylcholine in VLDL), **Methionine** (for choline synthesis via transmethylation), **Inositol**, Betaine, Vitamin B12, Folate. Deficiency → fat cannot be exported → accumulates.

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10 Memory Mnemonics

🧠 **MNEMONIC 1 — Β-OXIDATION 4 STEPS**

"Elephants Have Thick Hides" → EHTH

Enoyl-CoA dehydrogenase (FAD) → **H**ydratase (H_2O) → **T**hydroxyacyl-CoA dehydrogenase (NAD) → **T**hiolase (CoA) — products: $FADH_2$, NADH, Acetyl-CoA each round.

🧠 **MNEMONIC 2 — LIPOPROTEIN DENSITY ORDER**

"Chubby Vikings Inspire Lovely Humans"

Chylomicron → VLDL → IDL → LDL → HDL

Density increases (↑ protein, ↓ fat): Chylomicron = least dense, HDL = most dense.

🧠 **MNEMONIC 3 — KETONE BODY NAMES**

"A Beta-Blocker Acetates" → ABA

Acetoacetate · **B**eta-hydroxybutyrate · **A**cetone — three ketone bodies. Only first two carry energy; acetone is exhaled (fruity breath in DKA).

🧠 **MNEMONIC 4 — APOLIPOPROTEINS AND FUNCTION**

"A-I Activates LCAT; B-100 Binds LDL-R; C-II Calls LPL; E Everyone enters"

ApoA-I → activates LCAT (HDL maturation) · ApoB-100 → binds LDL receptor · ApoC-II → activates Lipoprotein Lipase (LPL) · ApoE → receptor-mediated uptake of remnants.

🧠 **MNEMONIC 5 — LIPOTROPIC FACTORS**

"Clever Mice In Big Labs"

Choline · **M**ethionine · **I**nositol · **B**12 · **L**ecithin — deficiency of any → fatty liver.

11 Clinical Correlations

🩺 **HIGH-YIELD CLINICAL LINKS**

- **Statin therapy:** Inhibit HMG-CoA reductase → ↓ cholesterol synthesis → upregulate hepatic LDL receptors → ↑ LDL clearance. Side effect: myopathy (↑ CK-MM). Monitor liver enzymes.
- **MCAD deficiency:** Medium Chain Acyl-CoA Dehydrogenase deficiency → β-oxidation block → hypoketotic hypoglycaemia in fasting infants → **SIDS-like presentation**. Increased C8-C10 acylcarnitines on newborn screen.
- **Alcohol → Fatty liver:** Ethanol → Acetaldehyde → NADH ↑ ↑ → NADH/NAD^+ ratio ↑ → OAA → Malate (depletes TCA) → FA synthesis ↑ + FA oxidation ↓ → TAG accumulates. Also: Acetaldehyde inhibits VLDL secretion.
- **Refsum's disease:** Phytanoyl-CoA hydroxylase deficiency → phytanic acid accumulates → peripheral neuropathy, retinitis pigmentosa, cerebellar ataxia. Avoid chlorophyll-rich foods.
- **Tangier disease:** ApoA-I deficiency → HDL deficiency → cholesterol accumulates in macrophages (foam cells) → orange tonsils, hepatosplenomegaly, ↑ atherosclerosis risk.
- **Abetalipoproteinaemia:** ApoB deficiency → cannot form chylomicrons + VLDL → fat malabsorption, acanthocytes, ataxia, retinitis pigmentosa. No dietary fat or fat-soluble vitamins absorbed.

🔥 VIVA RAPID FIRE — LIPID METABOLISM

▼ **Q1** **Why can't even-chain fatty acids contribute to net gluconeogenesis?**

> Even-chain FA → β-oxidation → Acetyl-CoA only. Acetyl-CoA cannot be converted back to pyruvate or OAA in net amounts because the Pyruvate Dehydrogenase reaction is irreversible. Acetyl-CoA entering TCA condenses with OAA → both carbons are released as CO_2 in the cycle — no net carbon gain for GNG. Exception: odd-chain FA → Propionyl-CoA → Succinyl-CoA → can enter GNG.

▼ **Q2** **Why does the liver produce but not use ketone bodies?**

> Liver lacks Succinyl-CoA transferase (3-oxoacid CoA transferase / thiophorase) — the enzyme needed for the first step of ketolysis. This is actually physiologically appropriate: the liver produces ketone bodies specifically to export fuel to the brain and other tissues during starvation. If the liver used its own ketones, this export function would fail.

▼ **Q3** **How does Malonyl-CoA coordinate FA synthesis and oxidation?**

> Malonyl-CoA is the first committed product of FA synthesis (made by ACC). It simultaneously inhibits CPT-I — the gatekeeper for FA entry into mitochondria for β-oxidation. So when FA synthesis is ON (high insulin, fed state), malonyl-CoA rises and blocks β-oxidation. When fasting → glucagon → ↓ malonyl-CoA → β-oxidation proceeds. One molecule coordinates two opposing pathways.

▼ **Q4** **Calculate ATP yield from complete oxidation of palmitate (C16:0).**

> Activation: −2 ATP (Acyl-CoA Synthetase). 7 rounds of β-oxidation: 7 $FADH_2$ (×1.5 = 10.5 ATP) + 7 NADH (×2.5 = 17.5 ATP) = 28 ATP. 8 Acetyl-CoA → TCA: 8 × 10 = 80 ATP. Total = 80 + 28 − 2 = **106 ATP** (modern P/O ratios). Using older textbook values (3 per NADH, 2 per $FADH_2$): 8×12 + 7×5 − 2 = 129 ATP. Match whichever your professor uses.

▼ **Q5** **What is reverse cholesterol transport and why is HDL "good"?**

> Reverse cholesterol transport is the process by which HDL picks up excess free cholesterol from peripheral tissues and transports it back to the liver for excretion as bile acids. Step 1: Nascent HDL (disc-shaped) secreted by liver/intestine. Step 2: ApoA-I activates LCAT → esterifies cholesterol → HDL matures (spherical). Step 3: Mature HDL delivers cholesterol ester to liver via SR-B1 receptor OR transfers via CETP to LDL/VLDL. This removes cholesterol from artery walls → anti-atherogenic = "good."

▼ Q6 **Why does alcohol cause fatty liver?**

> Ethanol → acetaldehyde (ADH) → acetate (ALDH). Both steps produce NADH. High NADH/NAD^+ ratio: (1) Inhibits β-oxidation (NAD^+ depleted). (2) OAA → Malate (TCA depleted → ↓ FA oxidation). (3) Acetaldehyde directly inhibits VLDL secretion. (4) ↑ Malonyl-CoA → ↑ FA synthesis. Net: FA cannot leave liver → TAG accumulates → fatty liver → if chronic, cirrhosis.

▼ Q7 **Distinguish between mitochondrial and peroxisomal β-oxidation.**

> Mitochondrial: substrate = C4–C20 FA; entry via carnitine shuttle; produces ATP directly (NADH + $FADH_2$ feed ETC); fully oxidises to Acetyl-CoA. Peroxisomal: substrate = VLCFA (>C20) and branched-chain FA; no carnitine shuttle needed; first oxidation uses FAD → H_2O_2 (neutralised by catalase); does NOT produce ATP directly — electrons go to O_2 not ETC; incompletely oxidises to medium-chain acyl-CoA for mitochondrial completion. Defect in peroxisomal → Zellweger, ALD.

▼ Q8 **Why is dipalmitoyl lecithin called the surfactant and what is its clinical significance?**

> Dipalmitoyl phosphatidylcholine (DPPC) has two saturated palmitate tails — this makes it pack tightly at the air-water interface of alveoli, dramatically reducing surface tension. By LaPlace's law, lower surface tension means smaller pressure required to keep alveoli open. Produced by Type II pneumocytes — develops fully only by ~35 weeks gestation. Premature birth before this → RDS (Hyaline Membrane Disease). Antenatal corticosteroids accelerate maturation. L:S ratio >2 in amniotic fluid = lung maturity.

13 Group-D EQ Template

📝 **HOW TO ANSWER: "HDL IS GOOD CHOLESTEROL" / "STATIN DRUGS ACT AS CHOLESTEROL-LOWERING AGENTS"**

Use the flowchart formula — Define → Mechanism → Kinetics/Regulation → Clinical consequence.

// EQ ANSWER FLOWCHART — "STATINS ACT AS CHOLESTEROL-LOWERING AGENTS"

Define
Competitive inhibitors of HMG-CoA reductase

↓

Mechanism
Statin → ↓ Mevalonate → ↓ Cholesterol synthesis in hepatocytes

↓

Compensatory upregulation
↓ Intracellular cholesterol → SREBP-2 activated → ↑ LDL receptor expression

↓

Clinical outcome
↑ LDL receptor → ↑ clearance of LDL from plasma → ↓ serum LDL

↓

Side effect
Myopathy (↑ CK-MM) — statin-induced myositis

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